

A Poor Programmer's Password Manager

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Abstract. You have some programming skills and use the Internet a lot. You don't want to be one of those unfortunates that gets their favourite password hacked, due to somebody's failure to properly protect the password file on a remote server. You don't want to change the world, you just want to work securely with the world as it is. You haven't the patience for elaborate procedures. You don't want it to cost you. How can you live with the broken Username/Password system, and yet feel certain that while the rest of the world may be vulnerable, you will be OK?

1 Introduction

First let us quickly recall how Username/Password works. The only cryptographic tool required is the humble one-way collision-free hash function, which can convert a string of any length into a unique fixed length (typically 256-bit) fingerprint. So as a function it works like $y = H(x)$ where x is a string, and y is the fingerprint. Inputs can be concatenated, so $y = H(a, b)$ just means that the string b is appended to the end of a to provide the input x . The one-wayness means that given just y , it's impossible to determine x . The collision free property means that it is computationally infeasible to find two x values which generate the same y .

Once you register as a client, the server generates a random "salt" s and stores $s, H(p, s)$ against your username, where p is your password. In practice various schemes have been proposed using different hash functions, and many incorporate iterative application of a fast hash function in order to artificially slow it down. All this information for every client is stored in a password file. When you enter your username and password, the server looks up your salt, calculates the hashed value, and compares it with the stored value. If it's good, you are in. Which is all very fine, but then a hacker steals the server's password file, and using a dictionary of common passwords, tries every one until they start getting hits, even if it takes them a substantial effort due to that artificial delay. Your only protection is to make sure your password is not in that dictionary. Hence those tedious password policies requiring you to remember an awful looking password for every service you use, and which you really need to write down somewhere.

For the hacker who has stolen the password file it's not personal. In fact they probably know nothing about you in advance of their attack.

2 A modest proposal

The proposed solution described here falls partially into the category of a “stateless” password manager. Passwords are not stored anywhere, they are generated afresh every time they are needed. It is not a new idea, see for example¹ and², and it has its rather strident critics - ³⁴⁵.

In our case the password manager is not rigidly stateless - it does support a database of non-secret meta-data associated with each password. For each service it stores and displays the associated website, the service name, the username, the policy index and a free-form field for any notes the client wishes to associate with the service.

But as regards the passwords themselves our password manager is stateless, as no passwords are stored. Its only fair to recall the objections to stateless password managers.

1. The password generated may not meet the website’s password policy
2. If the master password is compromised, all passwords are compromised, and each must be laboriously updated.
3. It is difficult to update an individual password, if that should be required.
4. If some existing password cannot be changed, then the password manager generated password cannot be used.
5. If one password is compromised, a brute force search could in theory reveal the master password.
6. If the password manager needs a bug-fix, then all passwords may need to be re-generated.
7. With a traditional encrypted password manager, two things are needed to obtain site-specific passwords: the ciphertext of the password vault, and the master password. With a stateless scheme, an attacker only needs one: the master password.
8. What if the website URL changes?

The twist proposed here is to use our programming skills to individualise our password manager, thus sidestepping most of these known shortcomings of classic stateless password managers. We will return later to respond to these objections.

3 Attacks and Defenses

First let us step back a bit and look at the issue from the hacker point of view. For a hacker the aim is to maximize the bang for the buck. In other words with

¹ <https://github.com/lesspass/lesspass>

² <https://vks.ai/ppm>

³ <https://security.stackexchange.com/questions/214301/what-are-the-cons-of-stateless-password-generators>

⁴ <https://kichik.com/2020/06/19/stateless-password-manager-usability/>

⁵ <https://www.sjoerdlangkemper.nl/2018/01/17/problems-with-pwdhash/>

the minimum amount of time and effort they would like to capture as many individual credentials as possible. Attacks can be classified as scalable or non-scalable. A scalable attack captures the credentials of multiple individuals with just one successful attack, and might be launched against the server as described above. A non-scalable attack on the other hand requires an individualized effort against each individual client.

One major weakness of the Username/Password mechanism is that the Username and the Password are transmitted directly to the server, albeit under cover of the TLS/SSL protocol. But this does imply that at both ends of the link there are potential points of attack, maybe by a virus, or by an employee on the server side. But a more likely possibility is that you have been drawn to a fake server that looks just like the real one. In that case when you entered your password, you just gave it away to be bad guys. In short you were the victim of a password phishing attack. To avoid such attacks a much better idea would be a zero-knowledge phishing-immune challenge-response based protocol in which the client secret is never transmitted from client to server, but rather used to construct a specific response to a random server challenge. Like Passkeys or MIRACL's M-Pin product. But that's another story.

For this exercise we will simply try and improve Username/Password as much as we can. Our plan is to avoid scalable attacks (a) on the server side by using very strong passwords, and (b) on the client side by deploying an individualized app, requiring a specific and targeted non-scalable attack to hack. We believe that just by forcing a hacker to go after individuals, rather than harvesting millions of passwords in a single attack, that this is enough to improve the user's security by many orders of magnitude.

4 Your Password Manager

Consider now a specially tailored password generator application. Stored somewhere is a random string, maybe created by the programmer getting their 3 year old grandson to bang on the keyboard

```
char r[]="dfgneNVriothj0erjgorigjq[89xu8qu1j)(kdUjSSgj4[jjjr";
```

We will refer to the amount of randomness or “unguessability” of a secret as its entropy. This is definitely a high entropy secret. Its one of those secrets which could not possibly be remembered and must be written down somewhere. So we write it down. Keep it in a file. When the program is run for the very first time the user is prompted to enter this secret. It will never be changed. Maybe a better idea would be one of those high entropy easier-to-type secrets like

```
char r[]="horse shit front door 8732";
```

Whatever you are happy with, as long as its high entropy. At a stroke this random string individualises your password manager and means that it is like no other.

The program consists primarily of a drop-down list with the names of the services with which the programmer has an account. There is also a box into which to enter the one-and-only Master Password p when the app first starts, and another box which will require a short 4-digit PIN number every time a password is needed.

The master secret should be one of those secrets which has at least 8 characters, but ideally with a healthy mix of upper and lower case alphanumerics and special characters. This secret should be remembered and NOT written down. But the good news is that now there is just one such password to be remembered. Finally there is a PIN number, the same to be used for all services. That's easy to remember.

When the name of a service n is selected from the drop-down list, the associated website URL and the Username for the service are displayed. The URL displayed is a safe place from which to navigate to the log-in page. If the URL has been entered to the browser from another source, it should at least be checked to ensure that it is consistent with the one displayed here – otherwise the user could be subject to a phishing attack. The Username is copied into the clipboard and can be pasted from there into the website username field.

Finally on entering the PIN a suitable password for the selected service is calculated as $pw_n = H(r, H(PIN, H(n, H(p))))$, and again copied to the device's clipboard, for pasting into the password field on the website. Observe that the password is generated from a hash of the random string, the master password, the PIN, and the user's chosen handle for the service (like "GitHub"). Note that by default the entered password pw_n is never visible, and so probably remains unknown and unremembered by the user (and to any shoulder-surfers). For the hash function $H()$ we use the latest standard SHA3-512 function.

We are assuming of course that the average user has no problem remembering one secure master password, as required by all password managers, and a PIN number.

The app should be compiled and copied to other computers and devices that the programmer owns.

5 Policy

To create the passwords the output of the final hash is converted to base64 encoding, which gives a nice mix of upper and lower case letters and digits, and occasional special symbols as dictated by most password policies. The default policy checks the output to ensure it contains at least one uppercase letter, one lowercase letter, one digit and one special character. If it doesn't it is hashed again until it does. Another decision is the length of the password. We decided on a default of 12 characters, giving roughly 72-bits of entropy. So a typical password generated by this program looks like `hSU!pKA09v6Z`. Normally the password is

just silently dropped into the device’s clipboard, but it can be made visible if desired, for example if it needs to be manually copied to an unsupported device.

Probably the best policy is policy 4. This is our recommended default. It avoids the ‘+’ symbol which is a problem for certain services, and converts ‘I’ to ‘i’ and ‘l’ to ‘L’, as these can look very similar if trying to manually enter the password. Policy 5 is a 16 character alternative to policy 4, which is useful for some rather paranoid website services.

One of the main criticisms of stateless password managers, is that the password generated may not fit with the policy of a particular website. Such policies are notorious for their variety. For example a website might set a maximum password length of just 10 characters, or it may (or may not) demand the inclusion of special characters. In practice our default password policy will work in most cases, but on occasion it may have to be tweaked. But for a programmer the solution is quite simple. There is a “policy” function which returns true or false. For every new website, a new policy *may* be required. However in our experience three or four such policy functions suffice, and a new policy is rarely required. Next to each website in the meta-data database is stored an index which identifies the appropriate policy. You can examine the source code to see the policies we found necessary to support.

6 Security

One of the biggest disincentives for a hacker is to have to go after each service user individually, and hack into their individual device. That is a major hassle, and is probably outside of their skill set. And one component of the password is the random r tucked into a file which is sitting next to your very own app, a copy of which is not available to anyone else on the planet.

Ideally your one-and-only master password should not be in any dictionary, certainly when combined with the PIN. One attack is to steal the password file from a server, then access your computer and try password and PIN guesses until there is a hit on the password file entry. So as for standard password hashing we make it more difficult by aggressively iterating the application of the hash function, and calculate the password as $pw_n = H(r, H(PIN, H(n, H(H(H(...H(p)...))))))$, with p hashed maybe tens of thousands of times, so that it takes milliseconds to calculate. Indeed since you only enter it once during the working day, you might afford to wait even a second or two.

If some-one sits down at your computer while the app is active, they cannot generate your passwords, as they don’t know the PIN. Note that nothing related to the master password or the PIN is stored by the application. Importantly, if the wrong master password and/or PIN is entered, no warning is given. The passwords generated will simply be the wrong ones. Also the implementation is completely deterministic – no cryptographically secure random number generator is required. Which is a good thing as they are notoriously hard to implement properly.

There is a case for making the PIN 6 digits rather than 4, as PIN guesses can be spread across multiple sites. You decide.

A potentially serious issue is how to replace a compromised password. The solution is simple, for example replace the service handle “GitHub” with “GitHub 2”, which is sufficient to generate a new unrelated password. The same idea can be used if you are a victim of one of those horrible password policies which require you to change your password every month.

7 Implementation

To make life easier it makes sense to use a tool like Qt⁶, with its nice Qt Creator tool. Now the same source code can be used to generate the app across a range of devices. Pay attention to the open source licensing arrangements, although if I read it right, and since its for personal use, its free.

New services can be added and deleted. For first time use there is the once-off tedious task of updating all of your existing poorly chosen passwords. There is a price to be paid for taking control of your own security. However passwords are typically context-dependent; not all passwords will be needed for all of your devices.

When the app is running it looks as shown below on all devices, just before the master password is entered. Our user interface is extremely simple, and lacks any artistic merit. You can probably do a lot better.

We have implemented the password manager on all the devices available to us – a windows PC, an Apple Mini Mac, a Linux laptop, an iPad and several iPhones of varying vintage. Implementation on the Apple devices, which requires the Xcode tool, was not entirely straightforward. One issue that arose was the necessity to pay for an Apple Developer license (\$99 per year) so that we could load the application onto our iPhones and iPads. We were frankly disappointed to have to pay anything for this project – but maybe you already hold such a license.

8 Operation

Operation is as simple and as user-friendly as we could make it. Enter the Master password on start-up, and press enter. Thereafter select a service, paste the username, enter the PIN, and paste the password.

8.1 Co-existing with the browser

Browsers go to great lengths to make password access as painless as possible (for an example see the appendix). Typically they all support their own in-built password managers. One big idea is to allow “syncing” of passwords across devices, using the browser’s own cloud resource. Syncing also means that if (when?) you

⁶ www.qt.io

lose your device, a replacement device can have its passwords restored from the cloud. But of course each different browser is going to do its own thing, each maintaining its own independent cloud resource. Third party password managers provide a similar browser-independent functionality, but since you are renting space in the cloud, they are going to charge you for that. It's their basic business model. Often they also store unencrypted meta-data associated with your accounts, like maybe even your browsing history.

But this does mean that your passwords (and any associated meta-data) get stored up in the cloud. And we don't want that. We are going to "sync" the old-fashioned way, by installing our own individual password manager on all of our devices (and all using the same random string).

Having done that, you may now want to completely disable those inbuilt browser password managers. Loss of a device is still not critical, as the password manager can be restored if necessary from your source code. Or you may decide to continue to trust your browser, in which case the passwords generated only need to be entered once, and thereafter the browser will remember them for you. A sensible compromise, that keeps most of the nice advantages of browser syncing (like autofilling of passwords), would be to insist that the browser keeps passwords locally and does not propagate them to any cloud, by tuning the sync settings such that everything *except* passwords continue to be synced across devices.

9 Objection responses

Finally we give our response to the objections listed above.

1. Simply program in a new policy
2. Compromise of a password manager's master password will always be bad news. But it is possible to make password guesses prohibitively expensive. And the attacker still needs to guess the PIN and access the random string. However there is no ducking the fact that the user does have a responsibility to come up with one good strong memorised password.
3. Simply add a counter to the website's handle and thus create a brand new utterly different password. It's a hassle, but any website that insists on rotating passwords is always going to be a hassle to use.
4. Such passwords should be handled externally to this password manager. It's a corner case we don't need to support.
5. Again make master password guesses expensive. And the attacker also needs to guess the PIN. And access the random string.
6. True. But this password manager is very simple, using just one cryptographic function. The code size is only about 500 lines (excluding the standard SHA3 code). But do think long and hard before deciding on and finalizing the individualization code tweaks.
7. Now an attacker needs 3 things, the app itself and its unique random string, the master password, and the PIN.

8. Not a problem as the URL is not included in the hash, only the user chosen website handle.

10 Conclusion

Having proposed this idea, there is really no excuse for me not to implement it for my own use. This I have done. Gradually I have migrated more services, requiring only a new entry into the app's internal database, and the occasional implementation of support for a new policy. I continue to allow browser syncing of data across my devices, but I have disabled password syncing. One way of looking at it - I am syncing passwords across my devices by loading my password manager app onto all of them, rather than by using the cloud, while remaining completely device/browser independent. So far the experience has not been too painful, and my own previously rather ad hoc, chaotic and stress inducing approach to password management is in the process of being phased out. I have reason to be confident that my interaction with internet services is now just a private arrangement between myself and the service, with the browser's role being limited to establishing a secure TLS-backed link that authenticates the service to me and encrypts our transaction.

Accessing passwords from the browser

Let's take Google Chrome as an example. Your passwords are stored both locally and in the cloud. To access what is stored in the cloud enter `passwords.google.com` into the address bar. To see your passwords you will need to *authenticate yourself to your browser*. To access what is stored locally go to settings and select "Autofill and Passwords", then "Password Manager". This time to see your passwords you will need to *authenticate yourself to your device*. Note that different passwords or biometrics are typically needed to authenticate for each of these scenarios.

MyPassMan

My Password Manager Reset

Master

Service

Website

Username

Note

Enter PIN Clear

Password ☐ Show

New Policy ☐ Add

Service

Delete Sure?

Fig. 1. The Password Manager app.